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Early post-fire regeneration in subalpine heathland and grassland in the Victorian Alpine National Park, south-eastern Australia

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Abstract Landscape fire (at the scale of square kilometres or more) is relatively rare in the alpine and subalpine environments of Australia. In early 1998, a major fire (the 'Caledonia Fire'), burnt approximately 35 000 ha, of which approximately 3000 hectares was subalpine heathland, grassland and wetland within the Victorian Alpine National Park. This fire was one of only three landscape-scale fires that have occurred anywhere in the treeless vegetation of the Victorian Alps in the past 100 years, the others being in 1939 and 1985. Monitoring of regeneration in subalpine vegetation commenced 3 weeks postfire. Sites were established in burnt grassland at Holmes Plain (1400 m a.s.l.) and burnt grassland and heathland at Wellington Plain (1480 m a.s.l.), and in unburnt grassland at both sites. In burnt grassland and heathland, the fire consumed much of the vegetation, leaving extensive areas of bare ground. The cover of dense vegetation declined from > 70% prefire, to approximately 15% immediately postfire. Bare ground at the Holmes and Wellington Plains sites ranged from 70% to 85% immediately postfire. By May 2000, approximately 2.5 years postfire, dense vegetation cover in grassland had increased to approximately 20%, and bare ground had decreased to an average of approximately 30%. In unburnt grassland, dense vegetation cover was generally > 95%, and the amount of bare ground less than 5%. The tussock-forming snow grasses resprouted vigorously following fire, and had flowered prolifically after 1 year. In heathland, most of the shrubs were incinerated, leaving close to 100% bare soil. Since then, a number of grasses and some dominant shrubs have resprouted vigorously, with some seedling regeneration. By May 2000, in heathland, bare soil was still > 50% and dense vegetation < 20%. Such ground cover conditions during this early postfire period were well below prefire levels, and well below the levels necessary to protect alpine soils from erosion. The Caledonia Fire has provided a rare opportunity to study ecological processes associated with postfire regeneration in treeless subalpine landscapes.

Key words: alpine, bare ground, biomass, Caledonia Fire, catchment protection, disturbance, fire regime, litter, vegetation cover.

INTRODUCTION

The natural fire regimes of the treeless high mountain vegetation of Australia are difficult to quantify, and the role of fire in the dynamics of alpine and subalpine vegetation in Australia is largely unknown. At the landscape scale (that of square kilometres or more), fire appears to be relatively rare in these environments (Kirkpatrick & Dickinson 1984; Leigh *et al.* 1987; Wimbush & Forrester 1988; Williams & Costin 1994). Given that few fires have occurred in alpine and subalpine vegetation, knowledge of post-fire ecology, especially initial impacts and early responses, in alpine and treeless subalpine vegetation is also scant.

Aboriginal people utilized the high country, but little is known about their use of fire there. Good evidence exists for fire-stick farming in lowlands surrounding high mountains, but there is little evidence for the extensive use of fire, or even the need to use fire extensively, in the treeless vegetation (Flood 1980). Prior to European settlement, landscape-scale fires are likely to have occurred in high mountain treeless vegetation in mainland Australia only once or twice per century (Leigh *et al.* 1987). European settlement in and around the high country commenced in the mid-19th century. Prescribed burning was used by pastoralists in the late 19th and early 20th centuries to improve the quality of green pick, and miners used fire to clear undergrowth to expose geological features (Costin

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1954; Carr & Turner 1959a; Lawrence 1999). These practices all increased the frequency of fire in alpine and subalpine vegetation (Pryor 1939; Stretton 1946; Raeder-Roitzsch & Phillips 1958; Leigh *et al.* 1987; Banks 1989; Kershaw & Strickland 1989; Lawrence 1999). On mainland Australia, prescribed fire in treeless subalpine and alpine vegetation was banned in the 1940s and 1950s, as its use was linked clearly to land degradation in the high mountain catchments (Australian Academy of Science 1957; Costin *et al.* 1959; Williams & Costin 1994).

In Victoria, there are few detailed data on the occurrence of fires and their effects on alpine and subalpine plant communities. Extensive fires in treeless vegetation are known to have occurred on two occasions in the 20th century. According to Carr & Turner (1959a) '. . . small fires spread onto the high plains from the neighbouring slopes prior to 1939, such as in 1926, but these fires were of minor importance'. In contrast, the State-wide fires of 1939 burnt approximately 80% of the snow gum country on the Bogong High Plains above approximately 1500 m a.s.l. (Carr & Turner 1959a). Although these fires burnt much treeless subalpine and alpine vegetation, that is, heathland, grassland, and wetland (Carr & Turner 1959a), we do not know the precise extent or severity of these fires. In 1985, a fire at Mount Buffalo burnt approximately 11 000 ha of lowland and forested foothill vegetation, and approximately 150 ha of treeless subalpine heathland, grassland and wetland (Wahren *et al.* 1999). The subject of the present paper is an extensive fire that occurred in the south-eastern parts of the Victorian Alpine National Park in January 1998: the 'Caledonia Fire'.

Subsequent to the last glaciation, Australian alpine environments have evolved in response to relatively small-scale disturbances (1 m^2–1 ha) such as localized frost heave, wind blasting and insect herbivory. Larger scale disturbances such as fire and extensive grazing by large vertebrate herbivores over square kilometres of country are rare or non-existent (Williams 1990; Williams & Costin 1994). Under such a native regime of small-scale disturbances, Australian alpine environments are characterized by a high cover of vegetation (normally close to 100%) and relatively little exposed bare soil (< 1–2%; Australian Academy of Science 1957; Costin 1958; Williams & Costin 1994). The removal of vegetation and litter by fire, and the consequent exposure of bare soil, has major implications in Australian alpine landscapes, because exposure of alpine soils is undesirable for soil conservation purposes, and subsequent regeneration to adequate soil cover is slow (Wahren *et al.* 1994; Williams & Costin 1994).

In the present paper, we summarize the early post-fire changes in vegetation and ground cover conditions that have occurred in grassland and heathland on Holmes and Wellington Plains, two of the larger high plains burnt in the Caledonia Fire (Fig. 1). We focus on the first 2 years post-fire because there have been no detailed, quantitative studies of early post-fire responses in Australian alpine landscapes following fires of such magnitude. The specific aim is to document the changes in vegetation cover, the amount of bare ground, and changes to gaps that have occurred in grassland and heathland over the first 2 years post-fire. Both communities are extensive in treeless high mountain environments in Australia. Within each, the amount of bare ground and both the quality and quantity of vegetation cover, and the nature of gaps created by the death of grass tussocks or shrubs, are critical determinants of both vegetation change and landscape condition in Australian high mountain landscapes (Wahren *et al.* 1994; Williams & Costin 1994).

Fire and study sites

The Caledonia Fire started on 31 December 1997, from an abandoned campfire on the Caledonia River, in mixed lowland eucalypt forest. 1 January 1998 was a day of extreme fire weather: the predicted forest fire danger index (Luke & McArthur 1978) for the nearby town of Heyfield was 50 (Tolhurst 1998). Fire weather and drought conditions at this time were broadly similar to those of other years in the past century when widespread fires occurred in Victoria: 1983, 1967, 1964, 1939 and 1926 (McCarthy & Tolhurst 2000). The fire burnt approximately 35 000 ha of the Carey Forest, of which approximately 22 000 ha was within the Alpine National Park, and approximately 3000 ha was treeless subalpine vegetation.

The landscape above approximately 1400 m is a mosaic of high ridges, wooded slopes, rolling grassy plains and wetlands. The treeless plains and surrounding snow gum (*Eucalyptus pauciflora*) woodlands have been grazed during summer and autumn by free ranging cattle since the late 19th century and cattle were present at the time of the fire. Sporadic grazing continued for 3 months post-fire, but grazing was excluded from the fire-affected area for the 1998–1999 and 1999–2000 seasons.

The fires were intense crown fires in both the eucalypt forests of the foothills and the subalpine snow gum woodlands. In the treeless vegetation, fire intensity was unknown, but field observations indicated that the fire at Wellington Plain was more intense and burnt more of the vegetation than at Holmes Plain. At Holmes and Wellington Plains, > 90% of the grassland, heathland and snow gum woodlands were burnt, and the subalpine bogs were also extensively burnt. This represents the most extensive fire in alpine and subalpine vegetation since 1939 (McCarthy & Tolhurst 2000).

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METHODS

Monitoring sites

At the end of January 1998, 3 weeks after the fire, monitoring sites were established at Holmes Plain (146°45′14′′E, 37°24′32′′S; 1400 m a.s.l.) and Wellington Plain (146°49′54′′E, 37°29′46′′S; 1480 m a.s.l.). These plains were selected because they were representative of the extensive subalpine high plains in this region of the Victorian Alps (Walsh *et al.* 1984, 1986) and, in the case of Holmes Plain, because two sites had been established in grassland in 1991 as part of a general monitoring program in the high country. At both sites, the soil was alpine humus soil, a well-structured organic loam that is acid, nutrient poor, and is typical of most alpine and subalpine grasslands and heathlands in mainland Australia (Williams & Costin 1994). One of these sites was burnt, the other was outside the fire perimeter; both sites were used for the present study. Extra sites were established in burnt heathland and grassland at both Holmes and Wellington Plains, and in an unburnt area of grassland at Wellington Plain, close to the burnt grassland. This unburnt grassland was within the fire perimeter. Using this network of sites, the changes in the burnt grassland over time could be compared with those in long-unburnt grassland. Unfortunately, there were no suitable unburnt heathland sites.

Estimation of ground cover and biomass

Point quadrats, sampled along permanent transects (Kent & Coker 1992), were used to record species composition and ground cover conditions. Transects were 10 m long with points 4 mm in diameter, and 20 cm apart (50 points per transect). This method has been widely used for many decades to estimate the cover of individual species and life forms (grasses, forbs and shrubs) in treeless vegetation in Australia (Carr & Turner 1959b; Wimbush & Costin 1979a,b,c; Wahren *et al.* 1994). Ground cover at each point was rated in terms of three cover classes (Wahren *et al.* 1994). Cover class 1 (CC1) is fixed litter or dense live vegetation, cover class (CC2) provides intermediate vegetation cover but the litter is loose, and cover class (CC3) consists of bare ground or sparse vegetation and/or thin (< 5 mm thick) loose litter. Live vegetation and litter were measured together in all cover classes, as both are important for soil protection in Australian alpine environments (Wahren *et al.* 1994).

In the Holmes Plain grassland, there were 40 transects (including 13 from 1991) distributed over an area of approximately 40 ha of burnt grassland. There were an additional 10 transects (six from 1991) in the unburnt grassland, over an area of approximately 2 ha; burnt and unburnt grassland sites were approximately 1 km apart. At Wellington Plain, there were 40 transects established in burnt heathland over an area of approximately 40 ha, and 10 transects in burnt grassland over an area of 2 ha. Subsequent to January 1998, the vegetation was remeasured along the transects in May 1998, January, May and December 1999, and late April/early May 2000. In December 1999, four transects were set up in a patch of unburnt grassland at Wellington Plain, 200 m from the burnt grassland; these were remeasured in May 2000.

Above-ground biomass was sampled in March–May 2000, using direct harvest from 10 0.25-m^2 quadrats distributed at random in both burnt and unburnt grassland of Holmes and Wellington Plains. Samples were separated in the field into standing vegetation and litter components. Biomass samples were not collected from heathland because of the likely damage to regenerating areas where there was at least 50% bare ground and small shrub seedlings.

Assessment of grass tussock basal area, intertussock gaps and prefire shrub cover

In addition to changes in bare ground and vegetation cover, we documented changes in the size and type of gap, both interdominant and within-canopy, in grassland and heathland. Such gaps are important determinants of ecological processes in these communities (Williams & Ashton 1988; Williams 1992). The area occupied by intertussock gaps and *Poa* tussock bases was estimated by line intercepts (Bonham 1989). Intercepted basal tussock length was measured by using a metal pin, 4 mm in diameter, placed vertically against the edge of the tussock. Ten randomly distributed 10-m long transects were sampled in both burnt and unburnt grassland at Holmes Plain, and in burnt grassland at Wellington Plain. As there were no prefire data of shrub cover in open heathland, this was estimated at Holmes Plain by using line intercepts to determine the diameter of burnt shrubs, based on the extent of burnt branches. This method could not be applied at Wellington Plain because the fire had incinerated most shrubs, leaving few or no stems.

Data analysis

Point quadrat data were explored using box and normal probability plots (Sokal & Rohlf 1995), which showed that all data met the assumptions for ANOVA. As the vegetation was sampled at the same sets of transects over time, we used repeated measures ANOVA (von Ende 1993). Mean intertussock gap size and mean tussock basal size, in burnt and unburnt grassland, were

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analysed by one-way ANOVA, whereas the biomass data were analysed by two-way (site × fire) ANOVA. All analyses were undertaken using SYSTAT 8.0 (SPSS, Chicago).

RESULTS

Immediate fire impact

The Caledonia Fire affected most vegetation and all plant communities of both Holmes and Wellington Plains. The main impact was removal of vegetative cover and litter, resulting in high levels of bare ground and little vegetation cover. In burnt grassland at Holmes Plain, the average amount of CC1 was 17%, and the average amount of CC3 was 76%. This was a substantial decrease in the quality of cover at Holmes Plain compared with the prefire state (Fig. 2a). In burnt grassland at Wellington Plain, the respective figures were 11 and 85% (Fig. 2b). In burnt heathland at Wellington Plain, all shrubs were incinerated, resulting in virtually no vegetation (CC1 = 3%) and an almost complete cover of bare ground (CC3 = 96%; Fig. 2b).

Post-fire regeneration

Changes in vegetation cover and biomass

In burnt grassland, both at Holmes and Wellington Plains, and in burnt heathland at Wellington Plain, CC1 increased significantly, and CC3 decreased significantly over the 2.5-year study period, with most of the changes occurring in the first 18 months post-fire. There was a significant change in both CC1 and CC3 between February 1998 and May 1999, but between May 1999 and May 2000, there was little or no significant change in the average value of either cover class across sites (Fig. 2a). Similarly, there were significant changes in the cover classes in both grassland and heathland at Wellington Plain between January 1998 and May 1999, but relatively little change between May 1999 and May 2000 (Fig. 2b). By May 2000, the average CC3 in burnt grassland was 35% at Holmes Plain (Fig. 2a) and 25% at Wellington Plain (Fig. 2b). The respective figures for CC1 were 23 and 36%. In burnt heathland at Wellington Plain, average CC3 in May 2000 was 53%, and average CC1 was 16% (Fig. 2b).

The post-fire increase in vegetation cover at both Holmes and Wellington Plains was primarily due to the rapid resprouting of snow grass tussocks and forbs (Figs 3a,4). The cover of *Poa* increased rapidly post-fire in both grassland and heathland. At Holmes Plain it was 84% by January 1999 (Fig. 3a); at Wellington Plain it was approximately 65% in grassland by May 1999, and 55% in heathland (Fig. 4a,b). *Poa* cover remained at or near these levels at both sites until May 2000. This cover was almost completely due to surviving tussocks resprouting post-fire, rather than seedling establishment and growth. The rapid resprouting of the snow grasses was accompanied by pyrogenic flowering, which commenced in the year following the fire and occurred in virtually all tussocks. Forbs also re-established rapidly from perennating subterranean organs or seedlings. Forb cover was approximately 40% by January 1999 at both grassland sites, and by December 1999 at the heathland site (Figs 3a,4). Most forb cover was provided by *Asperula gunnii, Craspedia* spp. and *Leptorhynchos squamatus*. Some, such as *Bracteantha subundulata* and *Podolepis robusta*, occurred in distinct patches of 0.5–3 m^2.

In contrast to the rapid regeneration of snow grasses and forbs, shrubs in the heathland of Wellington Plain were slow to regenerate (Fig. 4b). Shrub cover increased at approximately 7% per annum, such that by May 2000, shrub cover was 17%. Although the prefire shrub cover at this site is unknown, it is likely to have been at least 50%, as the area had been mapped as closed heathland by Walsh *et al.* (1984). Shrubs, including the dominants, whether seedlings or resprouts, had not flowered after 2.5 years.

The cover classes at the various prefire or unburnt grassland sites showed some intersite variation across the landscape and over time. In 1991 at Holmes Plain, at one site that was subsequently burnt in 1998, CC1 was 49% and CC3 was 21% (Fig. 2a). At the first site, total live vegetation cover was 80%, of which *Poa* provided three-quarters of the cover and forbs approximately one-quarter. The intertussock gaps at this site were generally occupied by forbs only, with little or no live grass foliage overlapping the gap, and only sparse litter within the gap. At the second site, unburnt by the Caledonia Fire, the respective CC1 and CC3 values in 1991 were 69 and 3%, and 73 and 5% in February

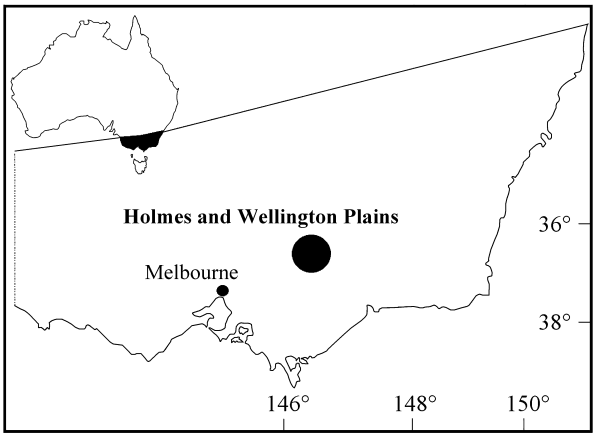


Fig. 1. Location map of Holmes and Wellington Plains.

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1998. At this site, CC1 increased significantly between February 1998 and May 1999 (Fig. 2a). Total *Poa* cover at this site was 95–99%, and forb cover 40–45%; shrub cover was consistently < 5% (Fig. 3b). The intertussock gaps at this site were relatively small (< 10 cm diameter) and well-covered by live *Poa* foliage and forbs. At the unburnt Wellington grassland site in December 1999, CC1 was 74% and CC 3 was 4% (Fig. 2b).

In May 2000, the biomass within burnt grassland was 8.6 t ha^{-1} at Holmes Plain, and 7.8 t ha^{-1} at Wellington Plain. Of this, an average of 5% was loose litter. The average biomass from burnt grassland at these two sites (8.2 t ha^{-1}) was significantly lower ($P < 0.001$) than that of unburnt grassland (15.8 t ha^{-1}). In unburnt grassland, 48% of the biomass was litter.

Gaps and canopies

In grassland, the average basal diameter of snow grass tussocks was significantly lower ($P < 0.001$) in burnt grassland (32 and 24 mm at Holmes and Wellington Plains, respectively) than in unburnt grassland (87 mm at Holmes Plain; Table 1). Inter-tussock gap size and relative area was significantly higher, by a factor of 2–3, in burnt grassland than in unburnt grassland.

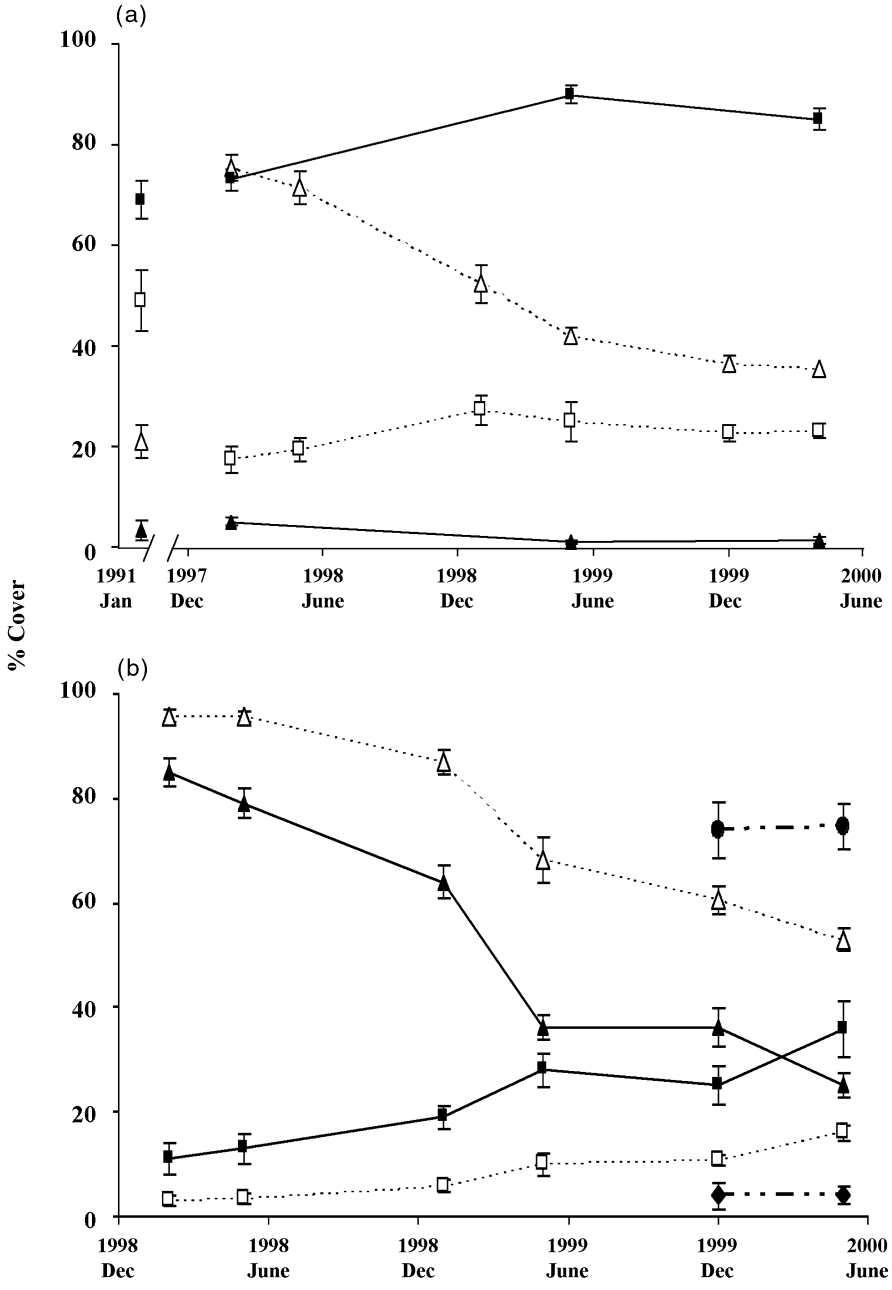


Fig. 2. Cover conditions on (a) grassland at Holmes Plain and (b) heathland and grassland at Wellington Plain. *y* axis = % Cover (± SEM; $n = 10$). (a) Cover class 1: (□), burnt; (■), unburnt. Cover class 3: (△), burnt; (▲), unburnt. (b) (■,□,●), Cover class 1; (▲,△,◆), cover class 3; (——), burnt grassland; (····), burnt heathland; (–··–), unburnt grassland.

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Table 1. Inter-tussock gap and tussock characteristics in burnt and unburnt grassland vegetation at Holmes and Wellington Plains, May 2000

	Gaps		Tussocks	
	Gap width (cm)	Area (%)	Basal diameter (mm)	Area (%)
Holmes burnt	44 ± 3.5	57	32 ± 2.7	43
Wellington burnt	38 ± 2.8	61	24 ± 1.9	39
Holmes unburnt	26 ± 2.1	24	87 ± 12.6	76

Mean (± SEM) width of intertussock gap, intercepted relative area of gaps, and mean base diameter of snow grass tussocks, and intercepted relative area of tussocks.

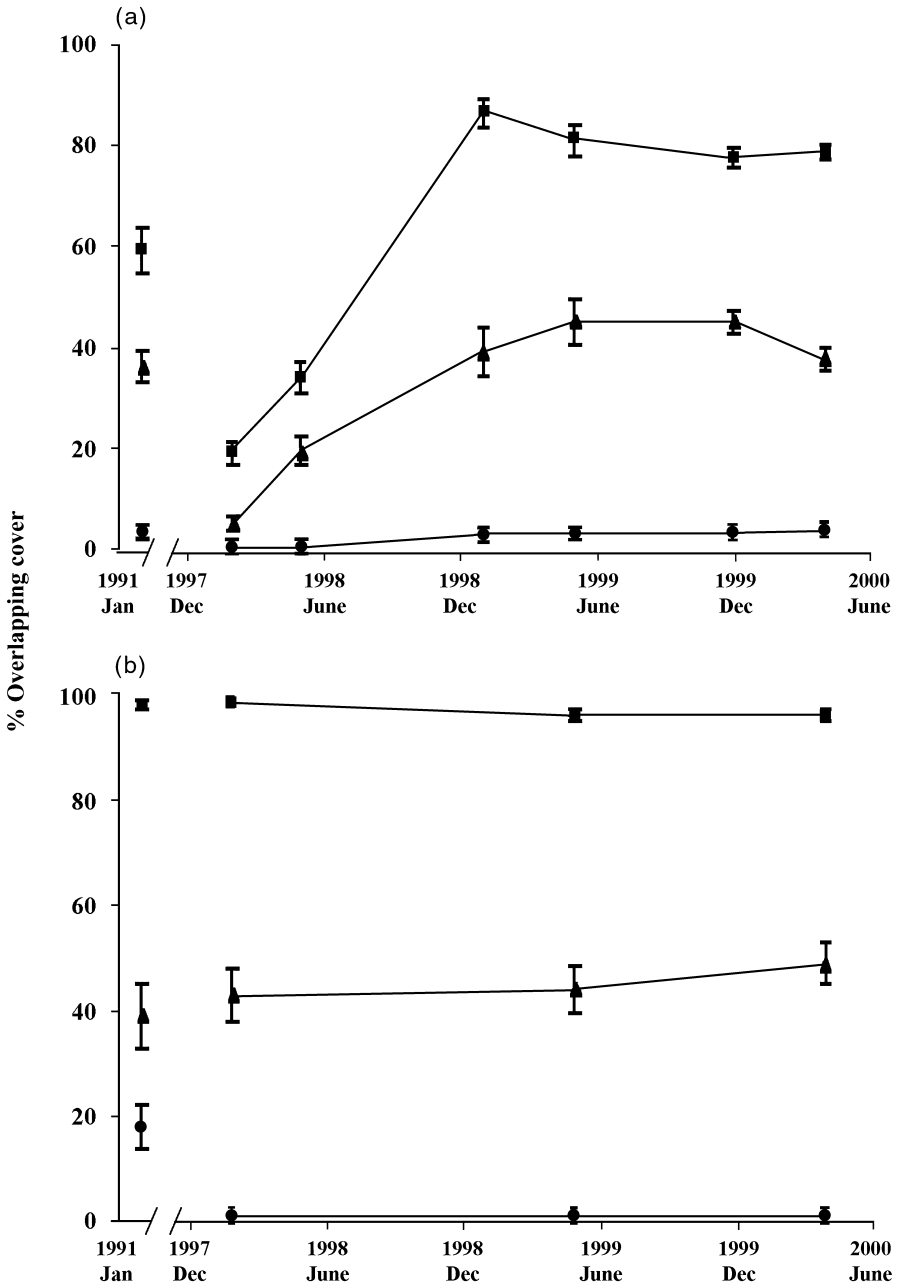


Fig. 3. Cover of main growth forms on Holmes Plain in 1991 and 1998–2000. (a) Burnt grassland; (b) unburnt grassland. *y* axis = % Cover (± SEM; $n = 10$). (■), *Poa*; (▲), forbs; (●), shrubs.

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In open heathland at Holmes Plain, the estimated prefire cover of *Grevillea australis* was approximately 20%. These *Grevillea* shrubs were killed and the litter beneath their canopies, which is normally thick (up to 5 cm) and loose, was completely burnt. The death of these shrubs has effectively altered the structure of this community on Holmes Plain from heathland to grassland. At Wellington Plain, where most shrubs were incinerated, the dominant shrubs, although resprouters, were still in the early stages of regeneration and yet to develop a full canopy. As for *Grevillea* at Holmes Plain, the litter associated with the shrubs at Wellington Plain was completely burnt.

DISCUSSION

Fire regimes in alpine and subalpine environments in Australia

The available historical, meteorological and biological evidence points strongly to landscape-scale fire being a relatively rare event in Australian alpine and subalpine vegetation. McCarthy and Tolhurst (2000) estimated that the fire-weather conditions associated with the Caledonia Fire have occurred five times in the past century. Assuming a relative lack of ignition sources in treeless high mountain vegetation compared with

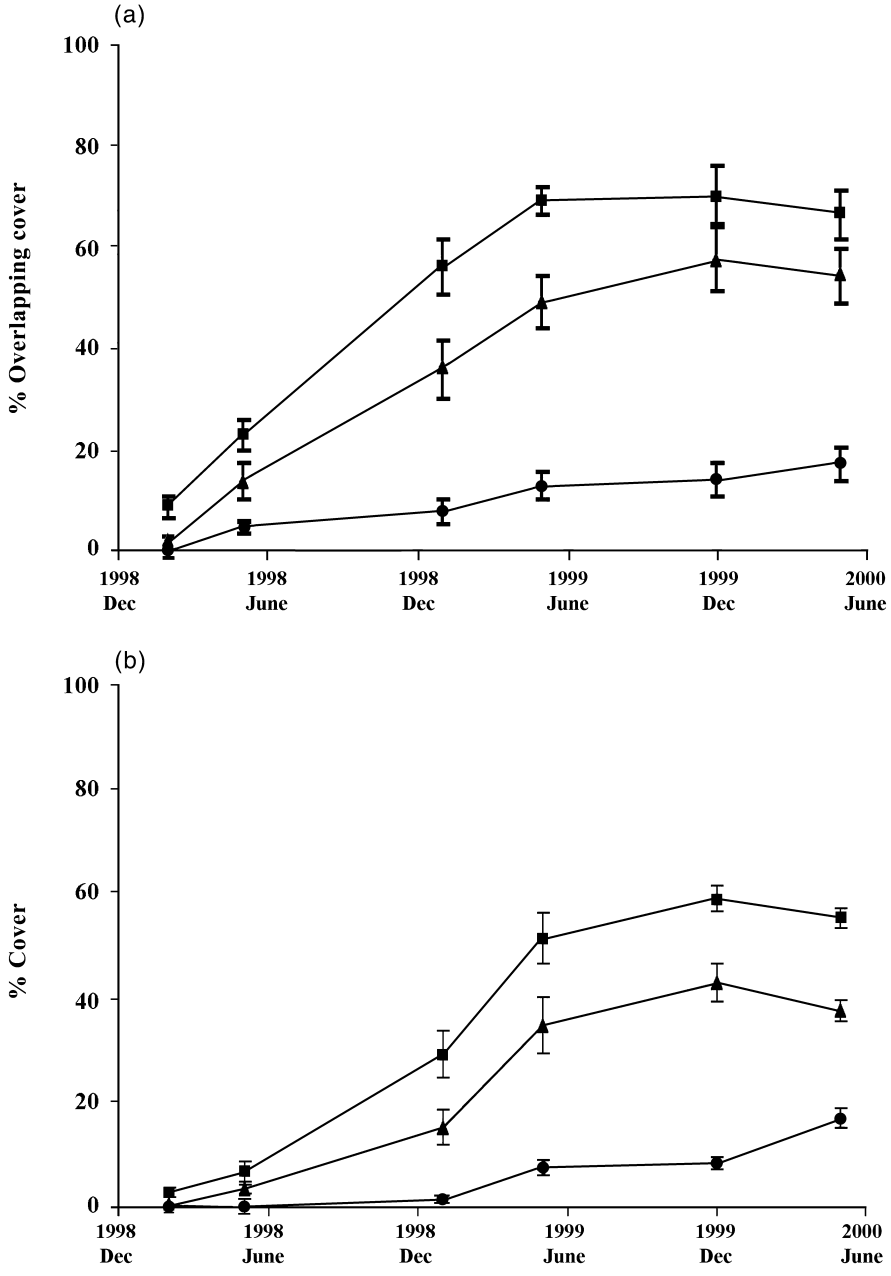


Fig. 4. Cover of main growth forms on Wellington Plains 1998–2000. (a) burnt grassland (b) burnt heathland. *y* axis = % Cover (± SEM; $n = 10$). (■), *Poa*; (▲), forbs; (●), shrubs.

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forest vegetation, they concluded that a fire frequency of once or twice per century was 'a reasonable prediction'. In subalpine areas in the ACT and NSW, the data of Pryor (1939) and Raeder-Roitzsch and Phillips (1958) are consistent with this conclusion.

This relatively low frequency of fire in Australian alpine landscapes is a function of the generally low to moderate fire weather conditions that occur at high altitude compared with landscapes at lower altitudes (Gill & Moore 1990). Fire in alpine and subalpine vegetation occurs only in exceptional circumstances, such as when drought is coupled with extreme fire-weather conditions. The fire discussed here has therefore provided a rare opportunity for detailed monitoring, in both the immediate post-fire period and the longer term, of the ecological processes governing post-fire vegetation change in high mountain treeless vegetation.

Post-fire patterns of regeneration

The most conspicuous response to fire was the rapid resprouting of most species. The vast majority of the alpine vascular flora has the capacity to resprout, from rhizomes or rootstock, and other perennating buds are generally protected by foliage and occur close to the ground (Williams & Costin 1994). This early onset of regeneration following fire is common in many ecosystems, and is often a function of increased soil nutrients in combination with a more open habitat and reduced competition (Gill 1981; Clarkson 1997; DeBano *et al.* 1998). In the Alps, such strong vegetative regeneration post-fire, however, is most likely an adaptation to the exigencies of the alpine climate rather than an adaptation to fire per se.

In the grasslands and heathlands, there was massed pyrogenic flowering of snow grasses, a phenomenon that is relatively common in the first year or two post-fire in alpine and subalpine grasses (Leigh *et al.* 1987; Yeates & Lee 1997). In the burnt grasslands and heathlands, such flowering will naturally provide a significant input to the soil seed-bank.

Based on 1991 figures, the cover of snow grasses and forbs had returned to approximately prefire levels by May 1999 (17 months post-fire). These responses occurred on both Holmes and Wellington Plain. This initial burst of regeneration was accompanied by a relatively rapid initial decline in the amount bare ground and/or thin loose litter with little vegetation cover (CC3) compared with the high values immediately post-fire.

Despite this rapid resprouting, and the return of total grass and forb cover to prefire levels within 2.5 years, the amount of CC3 (25–53%) was well above, whereas the amount of dense vegetation cover (CC1, 16–36%) was well below, levels typical for grassland and heathland in an unburnt state. At the majority of unburnt sites currently being monitored in the study area, the Holmes and Wellington Plains sites, and at independent sites at Wellington Plain and Mount Reynard (A. Tolsma, University of Melbourne School of Forestry, unpubl. data), vegetation cover was in excess of 95%, and bare ground was < 3%. Such values are typical of long unburnt, ungrazed alpine and subalpine vegetation elsewhere in Australia, where bare ground is generally less than 1–2%, and vegetation cover is almost complete (Costin *et al.* 1960; Wahren *et al.* 1994; Williams & Costin 1994; Williams *et al.* 1997; Wahren & Papst 2000).

While tussocks and shrubs establish and mature, the burnt grasslands and heathlands of these subalpine plains will remain vulnerable and sensitive to physical disturbances, such as frost heave, and any trampling by livestock. Post-fire regeneration studies from numerous alpine, subalpine or tundra sites around the world (Wimbush & Costin 1979a,b,c; Leigh *et al.* 1987; Racine *et al.* 1987; Gitay *et al.* 1992; Landhausser & Wein 1993; Wahren *et al.* 1994; Maumary *et al.* 1995; Williams *et al.* 1997; Yeates & Lee 1997; Vavrek *et al.* 1999) indicate that recovery to adequate levels of vegetation to protect the soil from subsequent disturbances is likely to take a decade or more. Given such high levels of bare or sparsely covered ground and low levels of vegetation cover in burnt sites, and the general slowness of vegetation regeneration following large-scale disturbances in Australian alpine environments, we conclude that the vegetation at Holmes and Wellington Plains has not recovered to prefire levels. Thus, there has been only limited post-fire regeneration, and, 2.5 years after the Caledonia Fire, the potential for soil erosion still exists in the burnt, treeless vegetation.

Patterns in grasslands

The biomass (including litter) of burnt grassland 2.5 years post-fire was significantly less than that of unburnt grassland, and less than that considered necessary for adequate protection of alpine soils ($10\ t\ ha^{-1}$; Costin *et al.* 1960). The litter component of biomass in burnt areas was negligible, whereas in the unburnt grassland, litter constituted almost half the biomass. Litter performs a crucial role in alpine environments, because it blankets the soils against extremes of temperature, and is a major source of organic matter that is an important determinant of soil chemical and physical health (Baldock & Nelson 2000). Such litter, much of which occurs in inter-tussock gaps, was totally consumed by the fire.

In grasslands, fire not only consumed litter, but resulted in a significant decline in the basal cover of snow grass tussocks and an increase in the size of inter-tussock gaps. By May 2000, the vegetation of these gaps was relatively sparse, with little dense cover. Thus, by

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increasing the size of the gaps, and consuming the litter within the gaps, fire has increased the potential area of sparse ground cover and therefore the potential for soil erosion. As regeneration in alpine communities is generally slow, the return of the litter component and the centripetal in-filling of gaps in the grasslands will take several more years.

Patterns in heathlands

Most shrubs were incinerated, as was the thick litter beneath the shrub canopy, causing almost 100% bare ground. Resprouting in the shrubs was slower than that of the herbaceous species. *Grevillea australis*, a shrub that is a known obligate seeder (Williams & Ashton 1988), was killed by 100% leaf scorch. Despite the decrease in the amount of bare ground in heathland post-fire, by May 2000 it was still high (mean 53%, pooling sites).

Whether or not a return to prefire state occurs will depend mainly on species, mode of regeneration and subsequent disturbance regime (e.g. Gill 1981; DeBano *et al.* 1998). Where *Grevillea* seedlings established, they were still < 10 cm tall 2.5 years post-fire, and development to mature shrubs will take a decade or more (Williams & Ashton 1987, 1988). Other post-fire regeneration studies in heathland elsewhere in the Victorian Alps, such as at Mount Fainter and Mount Buffalo (Wahren *et al.* 1999), have shown that recovery of heathland to prefire levels of vegetation cover (including litter) and bare ground is likely to take 10–15 years. As for the grasslands, then, the heathlands will therefore remain sensitive to physical disturbance while plants re-establish and mature.

Conclusion

The primary impact of fire in the subalpine landscapes on Holmes and Wellington Plain was, like any fire, the almost total removal of vegetation cover and litter and extensive exposure of bare ground. Such low levels of vegetation cover are highly unusual in virtually all unburnt Australian alpine and subalpine grassland and heathland communities. Although resprouting of herbaceous species following the fire was initially rapid in both grassland and heathland, the amount of vegetation cover in both communities 2.5 years post-fire was still insufficient to protect the soils. Soil and catchment protection is a major land use in the Australian Alps. As has been documented for other subalpine and alpine heathlands and grasslands subject to fire, the development of levels of vegetation and litter that will adequately protect the soil, and hence the catchments, will take a decade or more. The rarity of landscape-scale fire, and the slowness of regeneration post-fire, in treeless vegetation of the Australian Alps is such that the lands affected by the Caledonia Fire warrant intensive long-term studies. Such studies are the principal means of gaining an understanding of both species and plant community responses following major but rare disturbances such as a landscape-scale fire.

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